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(54) **POLYMORPHS OF TRIAZOLE ANTIFUNGAL COMPOUND PC945**

POLYMORPHE DER ANTIFUNGALEN TRIAZOLVERBINDUNG PC945

POLYMORPHES DE COMPOSÉ ANTIFONGIQUE DE TRIAZOLE PC945

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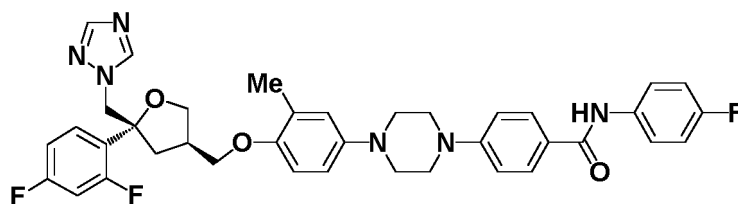
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(56) References cited:
WO-A1-2016/087878

• **THOMAS COLLEY ET AL: "In Vitro and In Vivo
Antifungal Profile of a Novel and Long-Acting
Inhaled Azole, PC945, on Aspergillus fumigatus
Infection", ANTIMICROBIAL AGENTS AND
CHEMOTHERAPY, vol. 61, no. 5, E02280, 21
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Compound I

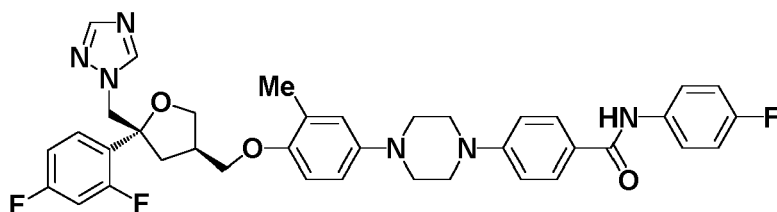
[0009] Compound I is a potent inhibitor of *Aspergillus fumigatus* sterol 14 α -demethylase (CYP51A and CYP51B) and has been characterised as a potent, long acting anti-fungal agent *in vitro* and *in vivo* (Colley *et al.*, 2017). Furthermore, it has been disclosed that topical treatment with Compound I, in combination with a known, systemic triazole agent, demonstrated synergistic antifungal effects against *Aspergillus fumigatus* in an *in vitro* human alveolus bilayer model and in the lungs of neutropenic, immunocompromised mice (Colley *et al.*, 2019). Compound I was also found to be a more potent inhibitor than posaconazole, voriconazole and fluconazole of *Candida auris* isolates collected globally (Shivaprakash *et al.*, 2019). The analysis of biomarkers, following intranasal dosing of Compound I in immunocompromised mice, showed that the antifungal effects accumulated upon repeat dosing and were persistent (Kimura *et al.*, 2017).

[0010] Polymorphism is an important consideration in the development of a medicinal product because of its influence on the physicochemical and biopharmaceutical properties of the compound. This characteristic arises whenever an entity, such as small molecule, can adopt different crystalline lattices. The resulting variation in solid state packing can lead to differences in storage stability, solubility profiles and density thereby affecting formulation and product manufacturing, as well as dissolution characteristics: an important factor in determining bioavailability *in vivo*.

[0011] There remains a need to provide polymorphic forms of Compound I for use in the treatment of mycoses. In particular, there is a requirement to provide Compound I in a crystalline form which has appropriate physical and chemical stability and other drug-related properties pertinent to its topical administration to the lung. These include its susceptibility to micronisation, its formulation as a readily dispersible, aqueous suspension, in addition to adequate bioavailability and acceptable pharmacokinetic properties.

Summary of the Invention

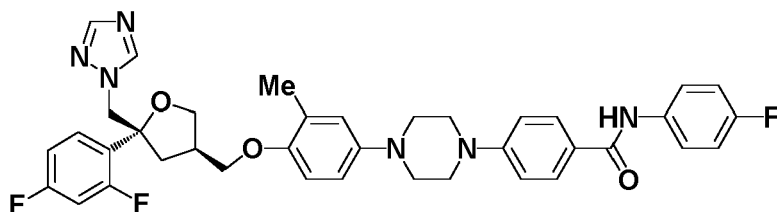
[0012] The invention provides a compound of formula (I):



Compound I

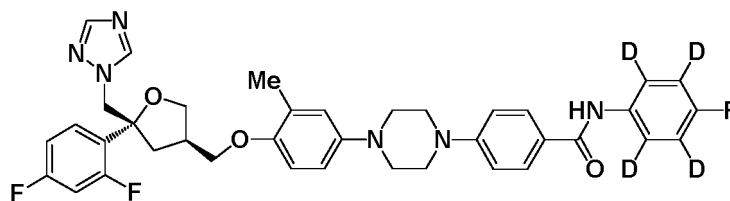
that is 4-(4-(4-(((3R,5R)-5-((1H-1,2,4-triazol-1-yl)methyl)-5-(2,4-difluorophenyl)tetrahydrofuran-3-yl)methoxy)-3-methylphenyl)piperazin-1-yl)-N-(4-fluorophenyl)benzamide in a crystalline form wherein the crystalline form is polymorphic Form 1.

[0013] In addition, the invention provides a compound of formula (I):



Compound I

that is 4-(4-(4-(((3R,5R)-5-((1H-1,2,4-triazol-1-yl)methyl)-5-(2,4-difluorophenyl)tetrahydrofuran-3-yl)methoxy)-3-methyl-

**Compound 2**

[0146] All instrument control, data collection, peak area integration and storage was performed using MassLynx (version 4.1). The mass spectrometer response (peak area ratio of analyte to internal standard) of each calibration standard was calculated by Watson LIMS (version 7.2) and plotted against the nominal (prepared) concentration. A weighted ($1/x^2$) least squares linear regression analysis was used to calculate an equation of the calibration curve. Concentrations of Compound I in the samples were back-calculated from the calibration curves to 3 significant figures. Microsoft Excel 2010 was used for further calculations. All percentages are presented to one decimal place.

[0147] Pharmacokinetic analysis was performed on Phoenix WinNonlin. Maximum mean plasma concentrations (C_{max}) and their times of occurrence (T_{max}) were determined. Areas under the mean plasma concentration-time curves up to the time of the last quantifiable sample (AUCt) were estimated by the linear trapezoidal rule. There were no adverse effects of the drug substance following its administration and serial blood samples were taken up to 48 hours after the start of the inhalation exposure.

[0148] The mean pharmacokinetic parameters for the two polymorphs following their inhalation administration to rats are summarised in **Table 16** and described in more detail below.

Table 16: Mean Pharmacokinetic Parameters for Forms 1 and 2 of Compound I Following their Inhalation Administration to Rats.

Pharmacokinetic Parameters	Mean	
	Form 1	Form 2
Nominal dose (mg/kg)	2.2	2.2
Estimated inhaled dose (mg/kg)	2.48	2.82
C_{max} (ng/mL)	4.95	8.59
T_{max} (hr)	6	6
AUCt (ng.h/mL) ^a	122	230
AUCt dose normalised (ng.h/mL) ^b	122	202
Relative bioavailability	Reference (100%)	166%
Footnotes: a. Both AUCt values are 0-48 hr; b. AUCt of Form 2 normalised to dose achieved for Form 1 (2.48 mg/kg)		

Plasma Concentrations

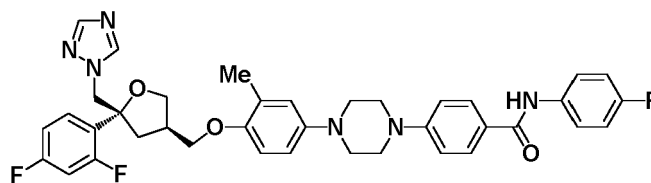
[0149] The plasma concentrations of drug following inhalation administration of Forms 1 and 2 remained quantifiable up to 48 hr after the start of exposure for both groups and are presented below (**Tables 17** and **18** respectively). The mean plasma concentration-time profiles derived therefrom are represented graphically (**Figs. 19** and **20** for Forms 1 and 2 respectively).

Table 17: Plasma Concentrations of Drug Following Inhalation Administration of Form 1 to Male Rats at a Nominal Dose Level of 2.2 mg/kg.

Time hr ^a	Drug Concentration (ng/mL) in Animal by No.						Mean	sd
	1	2	3	4	5	6		
IAD	1.16	0.826	1.35	-	-	-	1.11	0.27
3	-	-	-	5.31	4.35	2.41	4.02	1.48
4	4.73	6.32	3.78	-	-	-	4.94	1.28
6	-	-	-	5.38	3.97	5.49	4.95	0.85

Claims

1. A compound of formula (I):



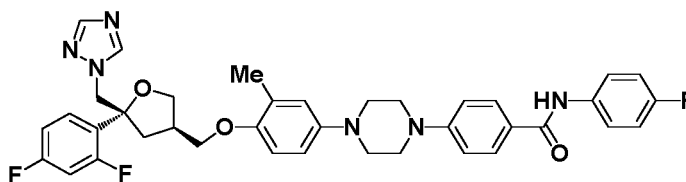
Compound I

that is:

4-(4-(4-(((3*R*,5*R*)-5-((1*H*-1,2,4-triazol-1-yl)methyl)-5-(2,4-difluorophenyl)tetrahydrofuran-3-yl)methoxy)-3-methylphenyl)piperazin-1-yl)-*N*-(4-fluorophenyl)benzamide in a crystalline form wherein the crystalline form is polymorphic Form 1, and wherein the crystalline form has:

- (a) an X-ray powder diffraction pattern containing seven peaks at (± 0.2) 7.0, 7.4, 7.9, 18.2, 19.7, 20.8 and 24.7 degrees 2-theta as measured using Cu K α radiation; and/or
(b) unit cell dimensions of 16.80 Å, 23.58 Å and 25.58 Å, and α , β and γ angles of 90°.

2. A compound of formula (I):



Compound I

that is:

4-(4-(4-(((3*R*,5*R*)-5-((1*H*-1,2,4-triazol-1-yl)methyl)-5-(2,4-difluorophenyl)tetrahydrofuran-3-yl)methoxy)-3-methylphenyl)piperazin-1-yl)-*N*-(4-fluorophenyl)benzamide in a crystalline form wherein the crystalline form is polymorphic Form 2, and wherein the crystalline form has:

- (a) an X-ray powder diffraction pattern containing six peaks at (± 0.2) 10.8, 17.0, 20.3, 22.7, 23.9 and 24.3 degrees 2-theta as measured using Cu K α radiation; and/or
(b) unit cell dimensions of 16.81 Å, 5.65 Å and 35.56 Å, and an α angle of 90°, a β angle of 101.54° and a γ angle of 90°.

3. The crystalline form of the compound according to claim 1 or claim 2 for use as a pharmaceutical in combination with a second or further active ingredient.
4. A pharmaceutical composition comprising the crystalline form of the compound according to claim 1 or claim 2, optionally in combination with one or more pharmaceutically acceptable diluents or carriers.
5. The pharmaceutical composition according to claim 4 wherein the crystalline form of the compound of claim 1 or claim 2 is in particulate form suspended in an aqueous medium.
6. The pharmaceutical composition according to claim 5 wherein the aqueous medium comprises at least 95% water, preferably at least 99% water.
7. The pharmaceutical composition according to any one of claims 4 to 6 wherein the aqueous medium comprises a charged tonicity adjusting agent, which is preferably sodium chloride.
8. The pharmaceutical composition according to any one of claims 4 to 7 wherein the aqueous medium comprises:

- (i) a co-solvent, preferably propylene glycol; and/or
- (ii) a surfactant, preferably Lipoid S100.

9. The pharmaceutical composition according to any one of claims 4 to 8 which comprises a second or further active ingredient.

10. The crystalline form of the compound for use according to claim 3 wherein the second or further active ingredient is selected from the group consisting of: anti-fungal agents (preferably voriconazole or posaconazole), amphotericin B, an echinocandin (preferably caspofungin), an inhibitor of 3-hydroxy-3-methyl-glutaryl-CoA reductase (preferably lovastatin, pravastatin or fluvastatin), candididin, filipin, hamycin, natamycin, nystatin, rimocidin, bifonazole, butoconazole, clotrimazole, econazole, fenticonazole, isoconazole, ketoconazole, luliconazole, miconazole, omoconazole, oxiconazole, sertaconazole, sulconazole, tioconazole, albaconazole, efinaconazole, epoxiconazole, fluconazole, isavuconazole, itraconazole, propiconazole, ravuconazole, terconazole, abafungin, amorolfine, butenafine, naftifine, terbinafine, anidulafungin, micafungin, benzoic acid, ciclopirox, flucytosine (5-fluorocytosine), griseofulvin, tolnaftate and undecylenic acid.

11. The crystalline form of the compound according to claim 1 or claim 2 or the pharmaceutical composition according to any one of claims 4 to 9 for use as a medicament.

12. The crystalline form of the compound according to claim 1 or claim 2 or the pharmaceutical composition according to any one of claims 4 to 9 for use in the treatment of mycoses or for use in the prevention or treatment of disease associated with mycoses.

13. The crystalline form of the compound or the pharmaceutical composition for use according to claim 12 wherein the mycosis is caused by:

- (i) *Aspergillus spp.*;
- (ii) *Aspergillus fumigatus* or *Aspergillus pullulans*, preferably *Aspergillus fumigatus*;
- (iii) an azole resistant *Aspergillus fumigatus*; or
- (iv) *Candida spp.*, preferably *Candida albicans* or *Candida glabrata*; *Rhizopus spp.*, preferably *Rhizopus oryzae*; *Cryptococcus spp.*, preferably *Cryptococcus neoformans*; *Chaetomium spp.*, preferably *Chaetomium globosum*; *Penicillium spp.*, preferably *Penicillium chrysogenum* or *Trichophyton spp.*, preferably *Trichophyton rubrum*.

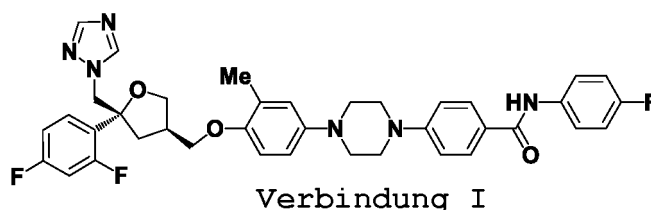
14. A process for preparing Compound I in crystalline form, wherein the crystalline form is polymorphic Form 2 according to claim 2, which comprises the steps of crystallising Compound I as polymorphic Form 2 from a solution of Compound I in a nominally dry organic solvent wherein the solvent is:

- (i) selected from the group consisting of MIBK, THF, PrOAc, EtOH, DME, MeOH, MeCN, MeNO₂, n-heptane, Et₂O, IPA, MEK, acetone, TBME and i-BuOH;
- (ii) a mixture comprising two or more solvents selected from the group consisting of MIBK, THF, PrOAc, EtOH, DME, MeOH, MeCN, MeNO₂, n-heptane, Et₂O, IPA, MEK, acetone, TBME and i-BuOH;
- (iii) a mixture of THF and toluene; or
- (iv) a mixture of THF and TBME.

15. A process according to claim 14, wherein the solvent is a mixture of THF and TBME.

Patentansprüche

1. Verbindung der Formel (I):

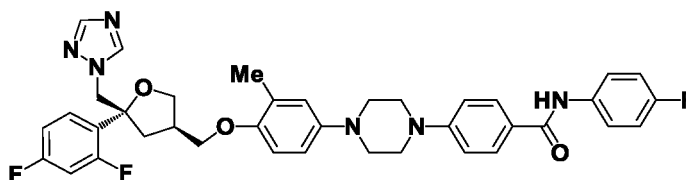


das heißt:

4-(4-(4-(((3*R*,5*R*)-5-((1*H*-1,2,4-Triazol-1-yl)methyl)-5-(2,4-difluorphenyl)tetrahydrofuran-3-yl)methoxy)-3-methylphenyl)piperazin-1-yl)-*N*-(4-fluorphenyl)benzamid in einer kristallinen Form, wobei die kristalline Form die polymorphe Form 1 ist und wobei die kristalline Form Folgendes aufweist:

- (a) ein Röntgenpulverbeugungsmuster, das sieben Spitzen bei ($\pm$ 0,2) 7,0, 7,4, 7,9, 18,2, 19,7, 20,8 und 24,7 Grad 2-Theta enthält, wie unter Verwendung von Cu K α -Strahlung gemessen wird; und/oder
- (b) Einheitszellabmessungen von 16,80 Å, 23,58 Å und 25,58 Å und α -, β - und γ -Winkel von 90°.

2. Verbindung der Formel (I):



Verbindung I

das heißt:

4-(4-(4-(((3*R*,5*R*)-5-((1*H*-1,2,4-Triazol-1-yl)methyl)-5-(2,4-difluorphenyl)tetrahydrofuran-3-yl)methoxy)-3-methylphenyl)piperazin-1-yl)-*N*-(4-fluorphenyl)benzamid in einer kristallinen Form, wobei die kristalline Form die polymorphe Form 2 ist und wobei die kristalline Form Folgendes aufweist:

- (a) ein Röntgenpulverbeugungsmuster, das sechs Spitzen bei ($\pm$ 0,2) 10,8, 17,0, 20,3, 22,7, 23,9 und 24,3 Grad 2-Theta enthält, wie unter Verwendung von Cu K α -Strahlung gemessen wird; und/oder
- (b) Zelleinheitsabmessungen von 16,81 Å, 5,65 Å und 35,56 Å, und einen α -Winkel von 90°, einen β -Winkel von 101,54° und einem γ -Winkel von 90°.

3. Kristalline Form der Verbindung nach Anspruch 1 oder Anspruch 2 zur Verwendung als ein Pharmazeutikum in Kombination mit einem zweiten oder weiteren Wirkstoff.

4. Pharmazeutische Zusammensetzung, umfassend die kristalline Form der Verbindung nach Anspruch 1 oder Anspruch 2, optional in Kombination mit einem oder mehreren pharmazeutisch unbedenklichen Verdünnungs- oder Trägerstoffen.

5. Pharmazeutische Zusammensetzung nach Anspruch 4, wobei die kristalline Form der Verbindung nach Anspruch 1 oder Anspruch 2 in partikulärer Form, die in einem wässrigen Medium suspendiert ist, vorliegt.

6. Pharmazeutische Zusammensetzung nach Anspruch 5, wobei das wässrige Medium mindestens 95 % Wasser, vorzugsweise mindestens 99 % Wasser, umfasst.

7. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 6, wobei das wässrige Medium ein geladenes Tonizitätseinstellungsmittel umfasst, welches vorzugsweise Natriumchlorid ist.

8. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 7, wobei das wässrige Medium Folgendes umfasst:

- (i) ein Co-Lösungsmittel, vorzugsweise Propylenglykol; und/oder
- (ii) ein Tensid, vorzugsweise Lipoid S100.

9. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 8, welche einen zweiten oder weiteren Wirkstoff umfasst.

10. Kristalline Form der Verbindung zur Verwendung nach Anspruch 3, wobei der zweite oder weitere Wirkstoff aus der Gruppe ausgewählt ist, die aus Folgendem besteht: antimykotische Mittel (vorzugsweise Voriconazol oder Posaconazol), Amphotericin B, ein Echinocandin (vorzugsweise Caspofungin), ein Inhibitor der 3-Hydroxy-3-methyl-

glutaryl-CoA-Reduktase (vorzugsweise Lovastatin, Pravastatin oder Fluvastatin), Candicidin, Filipin, Hamycin, Natamycin, Nystatin, Rimocidin, Bifonazol, Butoconazol, Clotrimazol, Econazol, Fenticonazol, Isoconazol, Ketoconazol, Luliconazol, Miconazol, Omoconazol, Oxiconazol, Sertaconazol, Sulconazol, Tioconazol, Albaconazol, Efinaconazol, Epoxiconazol, Fluconazol, Isavuconazol, Itraconazol, Propiconazol, Ravuconazol, Terconazol, Abafungin, Amorolfing, Butenafin, Naftifin, Terbinafin, Anidulafungin, Micafungin, Benzoesäure, Ciclopirox, Flucytosin (5-Fluorocytosin), Griseofulvin, Tolnaftat und Undecylensäure.

11. Kristalline Form der Verbindung nach Anspruch 1 oder Anspruch 2 oder pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 9 zur Verwendung als ein Arzneimittel.

12. Kristalline Form der Verbindung nach Anspruch 1 oder Anspruch 2 oder pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 9 zur Verwendung bei der Behandlung von Mykosen oder zur Verwendung bei der Prävention oder Behandlung von Erkrankungen, die mit Mykosen assoziiert sind.

13. Kristalline Form der Verbindung oder pharmazeutische Zusammensetzung zur Verwendung nach Anspruch 12, wobei die Mykose durch Folgendes verursacht wird:

(i) *Aspergillus* spp.;

(ii) *Aspergillus fumigatus* oder *Aspergillus pullulans*, vorzugsweise *Aspergillus fumigatus*;

(iii) einen Azol-resistenten *Aspergillus fumigatus*; oder

(iv) *Candida* spp., vorzugsweise *Candida albicans* oder *Candida glabrata*; *Rhizopus* spp., vorzugsweise *Rhizopus oryzae*; *Cryptococcus* spp., vorzugsweise *Cryptococcus neoformans*; *Chaetomium* spp., vorzugsweise *Chaetomium globosum*; *Penicillium* spp., vorzugsweise *Penicillium chrysogenum* oder *Trichophyton* spp., vorzugsweise *Trichophyton rubrum*.

14. Verfahren zur Herstellung von Verbindung I in kristalliner Form, wobei die kristalline Form die polymorphe Form 2 nach Anspruch 2 ist, welches die Schritte des Kristallisierens der Verbindung I als polymorphe Form 2 aus einer Lösung der Verbindung I in einem nominell trockenen organischen Lösungsmittel umfasst, wobei das Lösungsmittel Folgendes ist:

(i) ausgewählt aus der Gruppe bestehend aus MIBK, THF, PrOAc, EtOH, DME, MeOH, MeCN, MeNO₂, n-Heptan, Et₂O, IPA, MEK, Aceton, TBME und i-BuOH;

(ii) ein Gemisch, umfassend zwei oder mehr Lösungsmittel, die aus der Gruppe bestehend aus MIBK, THF, PrOAc, EtOH, DME, MeOH, MeCN, MeNO₂, n-Heptan, Et₂O, IPA, MEK, Aceton, TBME und i-BuOH ausgewählt sind;

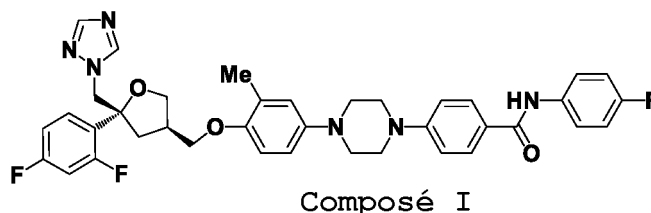
(iii) ein Gemisch aus THF und Toluol; oder

(iv) ein Gemisch aus THF und TBME.

15. Verfahren nach Anspruch 14, wobei das Lösungsmittel ein Gemisch aus THF und TBME ist.

Revendications

1. Composé de formule (I) :

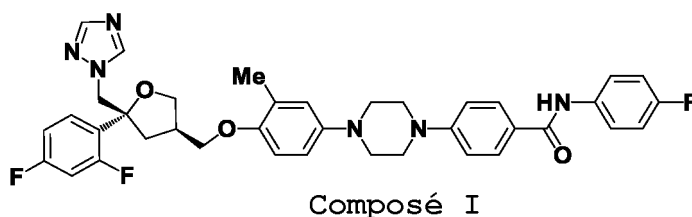


c'est-à-dire :

le 4-(4-(4-(((3*R*,5*R*)-5-((1*H*-1,2,4-triazol-1-yl)méthyl)-5-(2,4-difluorophényl)tétrahydrofuran-3-yl)méthoxy)-3-méthylphényl)pipérazin-1-yl)-*N*-(4-fluorophényl)benzamide sous une forme cristalline dans lequel la forme cristalline est la forme polymorphe 1, et dans lequel la forme cristalline a :

- (a) un diagramme de diffraction des rayons X sur poudre contenant sept pics à ($\pm 0,2$) 7,0, 7,4, 7,9, 18,2, 19,7, 20,8 et 24,7 degrés 2-thêta tel que mesuré en utilisant le rayonnement $K\alpha$ du Cu ; et/ou
 (b) des dimensions de la cellule unitaire de 16,80 Å, 23,58 Å et 25,58 Å, et des angles α , β et γ de 90°.

2. Composé de formule (I) :



c'est-à-dire :

le 4-(4-(((3*R*,5*R*)-5-((1*H*-1,2,4-triazol-1-yl)méthyl)-5-(2,4-difluorophényl) tétrahydrofuran-3-yl)méthoxy)-3-méthylphényl)pipérazin-1-yl)-*N*-(4-fluorophényl)benzamide sous une forme cristalline dans lequel la forme cristalline est la forme polymorphe 2, et dans lequel la forme cristalline a :

- (a) un diagramme de diffraction des rayons X sur poudre contenant six pics à ($\pm 0,2$) 10,8, 17,0, 20,3, 22,7, 23,9 et 24,3 degrés 2-thêta tel que mesuré en utilisant le rayonnement $K\alpha$ du Cu ; et/ou
 (b) dimensions de la cellule unitaire de 16,81 Å, 5,65 Å et 35,56 Å, et un angle α de 90°, un angle β de 101,54° et un angle γ de 90°.

3. Forme cristalline du composé selon la revendication 1 ou la revendication 2, pour une utilisation comme produit pharmaceutique en combinaison avec un second ou un autre ingrédient actif.

4. Composition pharmaceutique comprenant la forme cristalline du composé selon la revendication 1 ou la revendication 2, éventuellement en combinaison avec un ou plusieurs diluants ou supports pharmaceutiquement acceptables.

5. Composition pharmaceutique selon la revendication 4, dans laquelle la forme cristalline du composé de la revendication 1 ou de la revendication 2 est une forme particulière en suspension dans un milieu aqueux.

6. Composition pharmaceutique selon la revendication 5, dans laquelle le milieu aqueux comprend au moins 95 % d'eau, de préférence au moins 99 % d'eau.

7. Composition pharmaceutique selon l'une quelconque des revendications 4 à 6, dans laquelle le milieu aqueux comprend un agent d'ajustement de la tonicité chargé, qui est de préférence du chlorure de sodium.

8. Composition pharmaceutique selon l'une quelconque des revendications 4 à 7, dans laquelle le milieu aqueux comprend :

- (i) un cosolvant, de préférence du propylène glycol ; et/ou
 (ii) un tensioactif, de préférence le Lipoid S100.

9. Composition pharmaceutique selon l'une quelconque des revendications 4 à 8, comprenant également un second ingrédient pharmaceutique actif.

10. Forme cristalline du composé pour une utilisation selon la revendication 3, dans laquelle le second ou l'autre ingrédient actif est choisi dans le groupe constitué par : des agents antifongiques (de préférence le voriconazole ou le posaconazole), l'amphotéricine B, une échinocandine (de préférence la caspofungine), un inhibiteur de la 3-hydroxy-3-méthyl-glutaryl-CoA réductase (de préférence la lovastatine, la pravastatine ou la fluvastatine), la candidine, la filipine, l'hamycine, la natamycine, la nystatine, la rimocidine, le bifonazole, le butoconazole, le clotrimazole, l'éconazole, le fenticonazole, l'isoconazole, le kétoconazole, le luliconazole, le miconazole, l'omoconazole, l'oxiconazole, le sertaconazole, le sulconazole, le tioconazole, l'albaconazole, l'éfinaconazole, l'époxiconazole, le fluconazole, l'isavuconazole, l'itraconazole, le propiconazole, le ravuconazole, le terconazole, l'abafungine, l'amorolfine, la buténafine, la naftifine, la terbinafine, l'anidulafungine, la micafungine, l'acide benzoïque, le ciclopirox, la flucytosine (la 5-fluorocytosine), la griséofulvine, le tolnaftate et l'acide undécylénique.

Figure 1

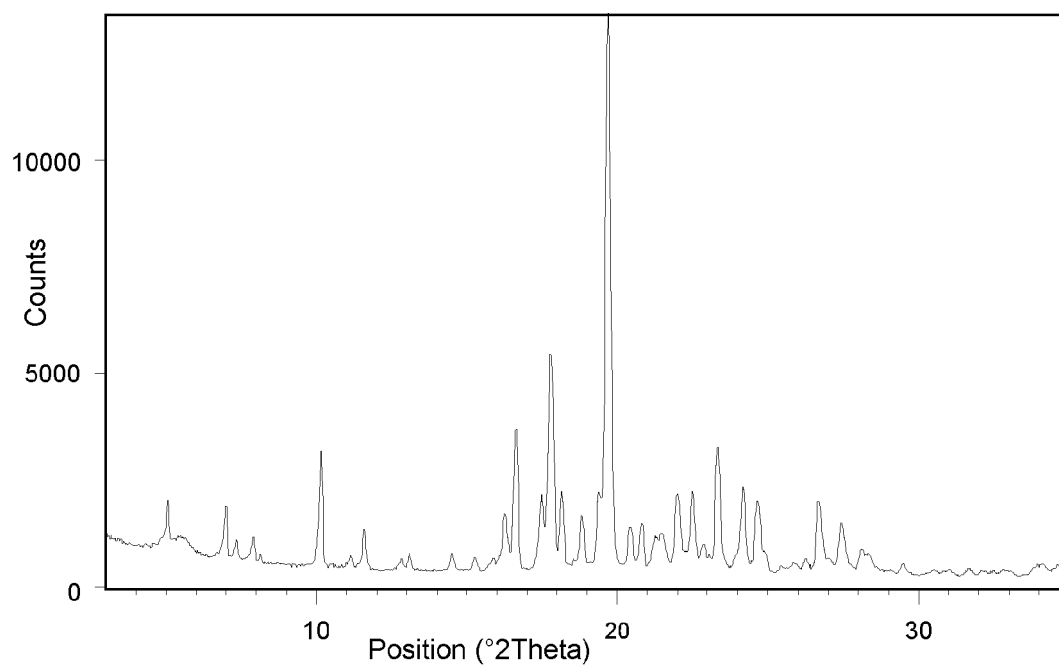


Figure 2

